

Abhishek Manish Lele

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Education

University of California San Diego, M.Eng. in Bioengineering (Medical Device Engineering) Sept 2025 – Dec 2026

- GPA: 3.74/4.0

- **Relevant Coursework:** Soft Robotics, Biomedical Integrated Circuits and Systems, Design and Control of Haptic Systems, Biomaterials & Medical Devices, Finite Element Methods I, Regulatory Affairs, Patient-Centric Clinical Medicine

Dr. Vishwanath Karad MIT World Peace University, B.Tech. in Mechanical Engineering Aug 2021 – May 2025

- GPA: 3.98/4.0

Work Experience

Mechanical Engineering Intern, Precima Technologies Pvt. Ltd. – Pune, India Jan 2025 – June 2025

- Developed 3D dental and maxillofacial implant models in SolidWorks from 2D specifications, incorporating realistic and worst case scenario boundary conditions and load paths as specified in ISO 14801.
- Conducted experimental fatigue testing of implants under cyclic loading, according to ISO 1099, using a Universal Testing Machine complied with ISO 7500. Performed Abaqus-based finite element simulations to predict fatigue life, achieving 93% agreement with experimental results.
- Iteratively refined geometry and modeling assumptions to improve simulation–experiment correlation to 96%. Optimized the implant thread geometry and the CNC toolpaths in Mastercam to improve manufacturability.

Biomechanics Research Intern, Deenanath Mangeshkar Hospital – Pune, India June 2024 – Dec 2024

- Acquired and analyzed kinematic, kinetic, and spatiotemporal gait data from Parkinson's patients using Vicon motion capture (100–200 Hz) and pressure platforms (1 kHz), extracting joint angles, plantar pressure distributions, ground reaction forces, and gait cycle features.
- Processed over 10,000 gait cycles collected per ISO 24227 to ensure statistically robust analysis of gait variability and spatiotemporal characteristics.

Academic Projects

AccuSense Cath: Soft-Tip Cardiac Catheter with Contact-Force Haptic Feedback Sep 2025 - Dec 2025

- Designed a 6 Fr cardiac ablation catheter with an elastic resin distal tip, Pebax braided shaft, and Nitinol flexure, complying with ISO 10993. Integrated a piezoresistive force-sensing system in the tip using CNT-doping and utilized deformation, measured by gage factor, contact angle, and speed to estimate applied force.
- Generated simulation data mapping input data to force, and trained a regression model for real-time estimation. Implemented real-time force estimation and haptic feedback using Arduino Nano 33 BLE and MATLAB, driving graded vibrotactile alerts via 3V coin motors.
- Modeled and prototyped using SolidWorks, ANSYS, and SLA 3D printing, achieving force sensing in the 0.02–0.4 N range with <50 ms latency.

TeleGlove: Bidirectional-Glove for Real-Time Hand Motion Tracking and Haptic Feedback Jul 2024 - Dec 2024

- Developed a wearable haptic glove integrating force-sensitive resistors (FSRs) at the fingertips, flex sensors for finger joint movement, and an IMU mounted on the dorsal wrist to capture hand position and finger-bending.
- Sensor data were fused to map position and motion within a real-time virtual environment implemented in Unity, enabling visualization and interaction such as grasping, squeezing, and throwing of virtual objects. The glove used vibrotactile motors at the outer fingertips to form a closed-loop system by providing a force-proportional haptic feedback.
- Achieved <100 ms virtual mapping latency and <150 ms end-to-end latency for haptic feedback from movement onset.

SoleSense: A Smart Insole for Foot Pressure Mapping and Orientation Analysis Jan 2024 - May 2024

- Designed a smart insole system instrumented with custom-fabricated pressure sensors and an inertial measurement unit (IMU) incorporated in a fPCB to capture plantar pressure distributions and foot orientation.
- Implemented real-time visualization in Processing, converting raw sensor data into gait features such as stance/swing phases, step timing, orientation. Used the insole in patient evaluations achieving 80% accuracy in foot orientation tracking during over-ground walking as compared to dynamic walkways.

Technical Skills

CAD/FEA Softwares: AutoCAD, SolidWorks, Creo, Fusion360, Catia, Ansys (Structural, Fluent), Abaqus, Mastercam

Human Motion and Biomechanical Data Acquisition Systems: FootWork, Vicon Nexus, OpenSim, LabChart

Programming & Visualization: MATLAB, Python, C++ , SQL, Arduino IDE, Processing, Unity